Prompt Engineering Primer: Four Practical Prompt Types

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Data science is one of the most important and fastest-growing fields in the modern digital world. Every day, enormous amounts of data are produced from mobilehones, social media, websites, online shopping platforms, banks, hospitals, satellites, and even smart home devices. This raw information is called data, but by itself it has no meaning. Data science is the process of collecting, organizing, cleaning, analyzing, and interpreting this data to extract useful knowledge and makeintelligent decisions. A person who performs these tasks is called a data scientist. A data scientist uses mathematics, statistics, computer programming, and logicalthinking to understand patterns hidden inside data. For example, when you watch videos on YouTube and the platform recommends similar videos you may like, it is data science working in the background. When Amazon or Flipkart suggests products based on your previous searches, it uses machine learning models created bydata scientists. Even Google Maps predicting traffic and suggesting the fastest route is an application of data science.The data science process usually starts with data collection. Data can be structured or unstructured. Structured data is organized in rows and columns like a spreadsheet or database table, while unstructured data includes images, audio, video, emails, and social media posts. After collecting the data, the next step is data cleaning. Real-world data often contains errors, missing values, duplicates, and noise. Cleaning the data is very important because wrong data will lead to wrong conclusions. Once the data is cleaned, data scientists perform exploratory data analysis. They create graphs, charts, and visualizations to understand trends and relationships. Visualization helps humans understand complex information easily. The next important stage is modeling. In this stage, data scientists build algorithms that can learn from data. These algorithms are part of machine learning. Machine learning allows computers to learn patterns without being explicitly programmed for every situation. There are different types of machine learning such as supervised learning, unsupervised learning, and reinforcement learning. In supervised learning, the computer is trained using labeled examples. For instance, to detect spam emails, the model is trained using emails already marked as spam or not spam. In unsupervised learning, the computer finds patterns without labels, such as grouping customers into categories based on purchasing behavior. Reinforcement learning is used when a system learns by trial and error, like robots learning to walk or AI learning to play chess

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Straightforward Prompts

Straightforward prompts are direct, natural-language questions designed to elicit clear, factual answers. They are best used for definitions or basic concept retrieval.

> **Prompt:** What is data science?
>
> *Answer:* It is the process of collecting, organising, cleaning, analysing, and interpreting data to extract useful knowledge.
>
> **Prompt:** What is machine learning?
>
> *Answer:* A technique where computers learn patterns from data without being explicitly programmed.

This method is highly effective when you require concise information without additional context or specific output formats.

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Tabular Format Prompting

Prompt	Response
What is Data Science?	Data science is the process of collecting, organizing, cleaning, analyzing, and interpreting data to extract useful knowledge and make intelligent decisions.
Who is a Data Scientist?	A data scientist is a person who uses mathematics, statistics, programming, and logical thinking to analyze data and find patterns.
Give an example of Data Science in daily life.	YouTube recommending videos, Amazon or Flipkart suggesting products, and Google Maps predicting traffic are examples of data science.
What is Structured Data?	Structured data is organized in rows and columns like spreadsheets or database tables.
What is Unstructured Data?	Unstructured data includes images, videos, audio, emails, and social media posts.
Why is Data Cleaning important?	Data cleaning removes errors, duplicates, and missing values so that analysis gives correct results.
What is Exploratory Data Analysis?	It is the process of using graphs, charts, and visualizations to understand trends and relationships in data.
What happens in the Modeling stage?	Data scientists build algorithms that learn patterns from data.
What is Machine Learning?	Machine learning allows computers to learn patterns from data without being explicitly programmed.
What are the types of Machine Learning?	Supervised learning, Unsupervised learning, and Reinforcement learning.

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Missing-Word Prompting

This technique, often called 'cloze' prompting, guides the AI to complete a specific thought or fill in a gap. It is an excellent way to test the model's understanding of terminology or to force a concise response.

"A person who analyses data and finds patterns is called a _____."

Answer: Data Scientist

"_____ recommends videos based on user viewing history."

Answer: YouTube

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Preceding-Question Prompting

1. What do we call the raw information produced from phones, websites, and social media?
   Answer: Data
2. What is the process of analyzing this data to extract useful knowledge called?
   Answer: Data Science
3. Who analyzes data using mathematics, statistics, and programming?
   Answer: Data Scientist
4. What do we call a person who finds patterns and insights from data?
   Answer: Data Scientist

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